

FC7xxx ICU Integration Manual

Rev.0.6

Revision History

Revision	Date	Changes
0.1	2023/07/14	Initial release for MCAL V0.1.0
0.6	2023/03/29	Updated for MCAL V0.6.0 - Added support for FC7240

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Chapter 1 Introduction

1.1 Introduction

This integration manual describes the integration requirements for the ICU module.

Chapter 2 Building

2.1 Dependencies on Other Modules

Module configuration dependency

- Common: This module is the basic module which used to choose the chip.
- EcuM: This module provides the wakeup source if wakeup source for any channel is configured.
- DET: This module provides the error detection functions during development.

Module build dependency

- Common: This module provides common type for other modules.
- Rte: This module provides APIs to protect/unprotect some parts of code from interrupts (Exclusive Areas).
- Det: This module is necessary for enabling Development error detection.

2.2 Files Required for Compile

ICU module files:

- MCAL/Src/Icu/Src/Icu_Aontimer.c
- _MCAL/Src/Icu/Src/Icu_Ftu.c
- _MCAL/Src/Icu/Src/Icu_Hw.c
- _MCAL/Src/Icu/Src/Icu_Port.c
- _MCAL/Src/Icu/Src/Icu_Cmp.c
- _MCAL/Src/Icu/Src/Icu.c
- _MCAL/Src/Icu/include/Icu_Aontimer_Types.h
- _MCAL/Src/Icu/include/Icu_Aontimer.h
- _MCAL/Src/Icu/include/Cmp_Reg.h
- _MCAL/Src/Icu/include/Icu_Cmp_Types.h
- _MCAL/Src/Icu/include/Icu_Cmp.h
- _MCAL/Src/Icu/include/Icu_Ftu_Types.h
- _MCAL/Src/Icu/include/Icu_Ftu.h
- _MCAL/Src/Icu/include/Icu_Hw_Types.h
- _MCAL/Src/Icu/include/Icu_Hw.h
- _MCAL/Src/Icu/include/Icu_MemMap.h
- _MCAL/Src/Icu/include/Icu_Port_Types.h
- _MCAL/Src/Icu/include/Icu_Port.h
- _MCAL/Src/Icu/include/Icu_Types.h
- _MCAL/Src/Icu/include/Icu_Version.h
- _MCAL/Src/Icu/include/Icu.h
- _MCAL_generate/src/Icu_PBcfg.c
- _MCAL_generate/src/Icu_Cfg.c
- _MCAL_generate/include/Icu_Cfg.h

Common module files:

- _MCAL/Src/Common/include/Mcal.h

- _MCAL/Src/Common/include/Std_Types.h
- _MCAL/Src/Common/include/Platform_Types.h
- _MCAL/Src/Common/include/Compiler.h
- _MCAL/Src/Common/include/Compiler_Cfg.h
- _MCAL/Src/Common/include/CompilerDefinition.h
- _MCAL/Src/Common/include/Ftu_Reg.h
- _MCAL/Src/Common/include/Port_Reg.h
- _MCAL/Src/Common/include/Aontimer_Reg.h
- _MCAL/Src/Common/include/Ftu_Common.h
- _MCAL/Src/Common/include/Ftu_RegOps.h

- _MCAL/Src/Common/src/Ftu_Common.c

Det module files:

- Det.h

Rte module files:

- SchM_Icu.h

2.3 Add Plug-ins

ICU module plug-ins are developed for EB tresos Studio, so, to use ICU plug-ins on the EB tresos Studio, the user needs to add the ICU module to the EB plug-ins folder first.

- 1) Copy the ICU module(_MCAL/EB_Plugins/eclipse/plugins/Icu) folder to EB tresos plug-ins (EB/tresos/plugins/) folder.
- 2) Set the ICU module output location folder for the generated source file and header files.
- 3) Use EB tresos to configure the ICU module and generate source and header files.

Chapter 3 Memory

3.1 Sections in Memory Map

Section Name	Section Type	Description
ICU_START_SEC_CONFIG_DATA_8 ICU_STOP_SEC_CONFIG_DATA_8 ICU_START_SEC_CONFIG_DATA_16 ICU_STOP_SEC_CONFIG_DATA_16 ICU_START_SEC_CONFIG_DATA_32 ICU_STOP_SEC_CONFIG_DATA_32	Variables	These are all the sections used for variables which have to be aligned to 8/16/32 bit. These variables are initialized by startup code (data).
ICU_START_SEC_CONFIG_DATA_UNSPECIFIED ICU_STOP_SEC_CONFIG_DATA_UNSPECIFIED	Configuration Data	Start and stop of Memory Section for Config Data.
ICU_START_SEC_CONST_BOOLEAN ICU_STOP_SEC_CONST_BOOLEAN ICU_START_SEC_CONST_8 ICU_STOP_SEC_CONST_8 ICU_START_SEC_CONST_16 ICU_STOP_SEC_CONST_16 ICU_START_SEC_CONST_32 ICU_STOP_SEC_CONST_32 ICU_START_SEC_CONST_UNSPECIFIED ICU_STOP_SEC_CONST_UNSPECIFIED	Variables	These are all the sections used for variables which have to be aligned to 8/16/32 bit or boolean. These variables are read only (rodata).
ICU_START_SEC_CODE ICU_STOP_SEC_CODE ICU_START_SEC_RAMCODE ICU_STOP_SEC_RAMCODE ICU_START_SEC_CODE_AC ICU_STOP_SEC_CODE_AC	Code	Start and stop of memory Section for Code(text).
ICU_START_SEC_VAR_NO_INIT_BOOLEAN ICU_STOP_SEC_VAR_NO_INIT_BOOLEAN ICU_START_SEC_VAR_NO_INIT_8 ICU_STOP_SEC_VAR_NO_INIT_8 ICU_START_SEC_VAR_NO_INIT_16 ICU_STOP_SEC_VAR_NO_INIT_16 ICU_START_SEC_VAR_NO_INIT_32 ICU_STOP_SEC_VAR_NO_INIT_32 ICU_START_SEC_VAR_NO_INIT_UNSPECIFIED ICU_STOP_SEC_VAR_NO_INIT_UNSPECIFIED	Variables	These are all the sections used for variables which have to be aligned to 8/16/32 bit. These variables are never cleared and never initialized by startup code (bss).
ICU_START_SEC_VAR_INIT_BOOLEAN ICU_STOP_SEC_VAR_INIT_BOOLEAN ICU_START_SEC_VAR_INIT_8 ICU_STOP_SEC_VAR_INIT_8	Variables	These are all the sections used for variables which have to be aligned to 8/16/32 bit. These variables are initialized by

Section Name	Section Type	Description
ICU_START_SEC_VAR_INIT_16		startup code (data).
ICU_STOP_SEC_VAR_INIT_16		
ICU_START_SEC_VAR_INIT_32		
ICU_STOP_SEC_VAR_INIT_32		
ICU_START_SEC_VAR_INIT_UNSPECIFIED		
ICU_STOP_SEC_VAR_INIT_UNSPECIFIED		

Chapter 4 Exclusive Area

ICU module using the services of Scheduler Module (SchM) for entering and exiting critical regions.

The following critical regions are used in the ICU driver:

- Icu.c:
 - ICU_EXCLUSIVE_AREA_00 Used in function Icu_SetBitChState to protect the set of the internal channel state
 - ICU_EXCLUSIVE_AREA_01 Used in function Icu_ClearBitChState to protect the clear of the internal channel state
 - ICU_EXCLUSIVE_AREA_02 Used in function Icu_StartTimestamp to protect the updates to:
 - Icu_aBuffer []
 - Icu_aBufferSize[]
 - Icu_aBufferNotify[]
 - Icu_aBufferIndex[]
 - Icu_aNotifyCount[]
 - ICU_EXCLUSIVE_AREA_03 Used in function Icu_SetActivationCondition to protect the updates to:
 - Icu_aChannelActivEdge []
 - ICU_EXCLUSIVE_AREA_04 Used in function Icu_TimeStampProcess to protect the updates to:
 - Icu_aBuffer []
 - Icu_aBufferIndex []
 - Icu_aNotifyCount []
 - ICU_EXCLUSIVE_AREA_05 Used in function Icu_GetTimeElapsed to protect the updates to:
 - Icu_aPeriod []
 - Icu_aActivePulseWidth[]
 - Icu_aPeriodOverflowCount[]
 - Icu_aActiveOverflowCount[]
 - ICU_EXCLUSIVE_AREA_06 Used in function Icu_GetDutyCycleValues to protect the updates to:
 - Icu_aPeriod []
 - Icu_aActivePulseWidth[]
 - Icu_aPeriodOverflowCount[]
 - Icu_aActiveOverflowCount[]
 - ICU_EXCLUSIVE_AREA_07 Used in function Icu_SignalMeasurementProcess to protect the updates to:
 - Icu_aSigMsmStart[]
 - Icu_aSignalMeasurementSeq[]
 - Icu_aActivePulseWidth[]
 - Icu_aPeriod[]
 - Icu_aSigMsmActivLevel[]
 - Icu_aSigMsmActive[]
 - Icu_aChannelOFCount[]
 - Icu_aActiveOverflowCount[]
 - Icu_aActivePulseWidth[]

- Icu_aChannelOFExCount[]
- ICU_EXCLUSIVE_AREA_08 Used in function Icu_StartSignalMeasurement to protect the updates to:
 - Icu_aActivePulseWidth []
 - Icu_aPeriod[]
 - Icu_aSignalMeasurementSeq[]
 - Icu_aSigMsmActive[]
 - Icu_aSigMsmStart[]
 - Icu_aChannelOFCount[]
 - Icu_aChannelOFExCount[]
 - Icu_aActiveOverflowCount[]
 - Icu_aPeriodOverflowCount[]
 - Icu_aChannelOverflowCount[]
- ICU_EXCLUSIVE_AREA_09 Used in function Icu_ResetEdgeCount to protect the updates to:
 - Icu_aEdgeCount []
- ICU_EXCLUSIVE_AREA_10 Used in function Icu_EnableEdgeCount to protect the updates to:
 - Icu_aEdgeCount []
- ICU_EXCLUSIVE_AREA_12 Used in function Icu_ChannelInterruptCallback to protect the updates to:
 - Icu_aEdgeCount []
- ICU_EXCLUSIVE_AREA_13 Used in function Icu_Cmp_GetInputState to protect the updates to:
 - Icu_aChannelOFCount []
 - Icu_aChannelOFExCount []
- ICU_EXCLUSIVE_AREA_ Used in function Icu_Cmp_SetNormalMode to protect the updates to:
 - ICU_CMP_HWA_SetIntEn_Rising[]
 - ICU_CMP_HWA_SetIntEn_Falling[]
 - ICU_CMP_HWA_ClearIntRisingFlag[]
 - ICU_CMP_HWA_ClearIntFallingFlag[]
- ICU_EXCLUSIVE_AREA_15 Used in function Icu_Cmp_GetInputState to protect the updates to:
 - ICU_CMP_HWA_ClearIntFallingFlag[]
 - ICU_CMP_HWA_ClearIntRisingFlag[]
- ICU_EXCLUSIVE_AREA_16 Used in function Icu_Cmp_SetActivationCondition to protect the updates to:
 - ICU_CMP_HWA_SetIntEn_Rising[]
 - ICU_CMP_HWA_SetIntEn_Falling[]
- ICU_EXCLUSIVE_AREA_17 Used in function Icu_Cmp_EnableEdgeDetection to protect the updates to:
 - ICU_CMP_HWA_ClearIntRisingFlag[]
 - ICU_CMP_HWA_ClearIntFallingFlag[]
- ICU_EXCLUSIVE_AREA_18 Used in function Icu_Cmp_DisableEdgeDetection to protect the updates to:
 - ICU_CMP_HWA_SetIntEn_Rising[]
 - ICU_CMP_HWA_SetIntEn_Falling[]
- ICU_EXCLUSIVE_AREA_19 Used in function Icu_Ftu_StopChannel to protect the updates to:
 - FTU channel control and status register

- s_aFtuChannelFlag[]
- ICU_EXCLUSIVE_AREA_20 Used in function Icu_Ftu_StartChannel to protect the updates to:
 - FTU channel control and status register
 - s_aFtuChannelFlag[]
- ICU_EXCLUSIVE_AREA_22 Used in function Icu_Ftu_EnableOverflow to protect the updates to:
 - s_aFtuOverflowFlag []
- ICU_EXCLUSIVE_AREA_23 Used in function Icu_Ftu_DisableOverflow to protect the updates to:
 - s_aFtuOverflowFlag []
- ICU_EXCLUSIVE_AREA_26 Used in function Icu_Ftu_SetActivationCondition to protect the updates to:
 - FTU channel control and status register
- ICU_EXCLUSIVE_AREA_27 Used in function Icu_Ftu_SetNormalMode to protect the updates to:
 - FTU channel control and status register
- ICU_EXCLUSIVE_AREA_28 Used in function Icu_Ftu_SetSleepMode to protect the updates to:
 - FTU channel control and status register
- ICU_EXCLUSIVE_AREA_29 Used in function Icu_Ftu_GetInputState to protect the updates to:
 - FTU channel control and status register
- ICU_EXCLUSIVE_AREA_31 Used in function Icu_Aontimer_GetEdgeNumbers to protect the updates to:
 - Aontimer counter register
- ICU_EXCLUSIVE_AREA_32 Used in function Icu_Aontimer_EnableEdgeDetection to protect the updates to:
 - Aontimer control status register
 - Aontimer compare register
- ICU_EXCLUSIVE_AREA_33 Used in function Icu_Aontimer_DisableEdgeDetection to protect the updates to:
 - Aontimer control status register
- ICU_EXCLUSIVE_AREA_34 Used in function Icu_Aontimer_SetNormalMode to protect the updates to:
 - Aontimer control status register
- ICU_EXCLUSIVE_AREA_35 Used in function Icu_Aontimer_SetSleepMode to protect the updates to:
 - Aontimer control status register
- ICU_EXCLUSIVE_AREA_36 Used in function Icu_Aontimer_SetActivationCondition to protect the updates to:
 - Aontimer control status register
- ICU_EXCLUSIVE_AREA_37 Used in function Icu_Aontimer_ResetEdgeNumbers to protect the updates to:
 - Aontimer control status register
- ICU_EXCLUSIVE_AREA_38 Used in function Icu_Aontimer_EnableEdgeCount to protect the updates to:
 - Aontimer control status register
 - Aontimer compare register
- ICU_EXCLUSIVE_AREA_39 Used in function Icu_Aontimer_DisableEdgeCount to protect the updates to:
 - Aontimer control status register
- ICU_EXCLUSIVE_AREA_40 Used in function Icu_Cmp_EnableEdgeCount to protect the updates to:
 - ICU_CMP_HWA_SetIntEn_Rising[]
 - ICU_CMP_HWA_SetIntEn_Falling[]

- ICU_EXCLUSIVE_AREA_41 Used in function Icu_Cmp_DisableEdgeCount to protect the updates to:
 - ICU_CMP_HWA_ClearIntRisingFlag[]
 - ICU_CMP_HWA_ClearIntFallingFlag[]
- ICU_EXCLUSIVE_AREA_42 Used in function Icu_Cmp_SetSleepMode to protect the updates to:
 - ICU_CMP_HWA_SetIntEn_Rising[]
 - ICU_CMP_HWA_SetIntEn_Falling[]
- ICU_EXCLUSIVE_AREA_59 Used in function Icu_Port_ProcessInterrupt to protect the updates to:
 - Port interrupt status flag register
- ICU_EXCLUSIVE_AREA_60 Used in function Icu_Port_ChannelStart to protect the updates to:
 - Port control register
 - s_aPortMask[]
- ICU_EXCLUSIVE_AREA_62 Used in function Icu_Port_SetActivationCondition to protect the updates to:
 - Port control register
- ICU_EXCLUSIVE_AREA_63 Used in function Icu_Port_GetInputState to protect the updates to:
 - Port control register
- ICU_EXCLUSIVE_AREA_64 Used in function Icu_Port_ChannelStop to protect the updates to:
 - Port control register
 - s_aPortMask[]

Chapter 5 Interrupt Service Routine (ISR)

Instance	Interrupt Name	IRQ Number (NVIC Interrupt ID)
FTU0	FTU0_IRQHandler	116
FTU1	FTU1_IRQHandler	117
FTU2	FTU2_IRQHandler	118
FTU3	FTU3_IRQHandler	119
FTU4	FTU4_IRQHandler	120
FTU5	FTU5_IRQHandler	121
FTU6	FTU6_IRQHandler	122
FTU7	FTU7_IRQHandler	123
FTU8 (For FC7300)	FTU8_IRQHandler	124
FTU9 (For FC7300)	FTU9_IRQHandler	125
FTU10 (For FC7300)	FTU10_IRQHandler	126
FTU11 (For FC7300)	FTU11_IRQHandler	127
PORTA	PORTA_IRQHandler	75
PORTB	PORTB_IRQHandler	76
PORTC	PORTC_IRQHandler	77
PORTD	PORTD_IRQHandler	78
PORTE	PORTE_IRQHandler	79
PORTF (For FC7300)	PORTF_IRQHandler	80
PORTG (For FC7300)	PORTG_IRQHandler	81
PORTH (For FC7300)	PORTH_IRQHandler	82
PORTI (For FC7300)	PORTI_IRQHandler	83
AONTIMER0	AONTIMER_IRQHandler	71
CMP0	CMP0_IRQHandler	128
CMP1	CMP1_IRQHandler	129
CMP2(For FC7300)	CMP2_IRQHandler	130

Chapter 6 Error Report

6.1 Det

Function Name	Error Type
Icu_Init	ICU_E_ALREADY_INITIALIZED ICU_E_PARAM_POINTER
Icu_DeInit	ICU_E_UNINIT
Icu_SetMode	ICU_E_UNINIT ICU_E_BUSY_OPERATION
Icu_DisableWakeup	ICU_E_UNINIT ICU_E_PARAM_CHANNEL
Icu_EnableWakeup	ICU_E_UNINIT ICU_E_PARAM_CHANNEL
Icu_CheckWakeup	ICU_E_UNINIT
Icu_SetActivationCondition	ICU_E_UNINIT ICU_E_PARAM_ACTIVATION ICU_E_PARAM_CHANNEL
Icu_DisableNotification	ICU_E_UNINIT
Icu_EnableNotification	ICU_E_UNINIT
Icu_GetInputState	ICU_E_UNINIT ICU_E_PARAM_CHANNEL
Icu_StartTimestamp	ICU_E_UNINIT ICU_E_PARAM_CHANNEL ICU_E_PARAM_POINTER ICU_E_PARAM_NOTIFY_INTERVAL
Icu_StopTimestamp	ICU_E_UNINIT ICU_E_PARAM_CHANNEL ICU_E_NOT_STARTED
Icu_GetTimestampIndex	ICU_E_UNINIT ICU_E_PARAM_CHANNEL
Icu_ResetEdgeCount	ICU_E_UNINIT ICU_E_PARAM_CHANNEL
Icu_DisableEdgeCount	ICU_E_UNINIT ICU_E_PARAM_CHANNEL
Icu_GetEdgeNumbers	ICU_E_UNINIT ICU_E_PARAM_CHANNEL
Icu_EnableEdgeDetection	ICU_E_UNINIT ICU_E_PARAM_CHANNEL
Icu_DisableEdgeDetection	ICU_E_UNINIT ICU_E_PARAM_CHANNEL
Icu_StartSignalMeasurement	ICU_E_UNINIT

Function Name	Error Type
	ICU_E_PARAM_CHANNEL
Icu_StopSignalMeasurement	ICU_E_UNINIT ICU_E_PARAM_CHANNEL
Icu_GetTimeElapsed	ICU_E_UNINIT ICU_E_PARAM_CHANNEL
Icu_GetDutyCycleValues	ICU_E_UNINIT ICU_E_PARAM_CHANNEL
Icu_GetInputLevel	ICU_E_UNINIT ICU_E_PARAM_CHANNEL

6.2 Dem

None

Chapter 7 Function Calls

7.1 Function Calls during Startup

The API needs to be called is `Icu_Init(&ConfigPtr);`

7.2 Function Calls during Shutdown

None

7.3 Function Calls during Wake-up

None

7.4 Function Calls during Runtime

None

Chapter 8 Other Requirements

8.1 Calls to Notification Functions, Callbacks, Callouts

If the user configures notifications and the value is not "NULL_PTR" or "NULL", an extern declaration will generate in Icu_Cfg.c or Icu_PBCfg.c. User need implement the notification in any file.

8.2 Macros

Please check various definitions available in Common module's include file Mcal.h for details.

- If OS is not used:

AUTOSAR_OS_NOT_USED needs to be defined.

In this case, If USE_SW_VECTOR_MODE is not defined, the user needs to map the interrupt vector table function to ISR function, for example:

```
extern ISR(ICU_PORT_A_ISR_USED);

void PORTA_IRQHandler(void)

{

    ICU_PORT_A_ISR_USED();

}
```

- If OS is used:

Do not define AUTOSAR_OS_NOT_USED.

Chapter 9 Integration Steps

- 1) Configure the ICU module and generate configuration files (please refer to Building chapter for details).
- 2) Configure appropriate memory sections in linker file or other (please refer to Memory chapter for details).
- 3) Map interrupt notification to their vector locations (please refer to chapter ISR for details).
- 4) Build the ICU module with dependent modules.